

Worksheet answers**Part 1: Primate trends**

Anatomy and behaviour are important tools that scientists use to determine how organisms are related. Compare five primate skulls using the learning object, *Primate trends*. As you select skulls, rate each characteristic, and use this information to determine which are most closely related.

- On the 'Meet the primates' screen, of the learning object, compare the five skulls and list them in order from smallest to largest.

gibbon, chimpanzee, human, orang-utan, gorilla

- Complete the table below as you work through *Primate trends*. In each column, list names of primates in order (top to bottom) from smallest to largest (or shortest to longest) skull feature. Change skull type by selecting the button at the top left of the window showing the skull you want to view.

SKULL FEATURE				
cranial capacity	canine length	size of mandible relative to skull	size of brow ridge	length of maxilla
<i>gibbon</i>	<i>human</i>	<i>gibbon</i>	<i>homo</i> (no brow ridge)	<i>gibbon</i>
<i>chimpanzee & orang-utan</i>	<i>gibbon</i>	<i>human</i>	<i>orang-utan</i>	<i>human</i>
	<i>chimpanzee</i>	<i>gorilla</i>	<i>gibbon</i>	<i>chimpanzee</i>
<i>gorilla</i>	<i>orang-utan</i>	<i>orang-utan</i>	<i>chimpanzee & gorilla</i>	<i>gorilla & orang-utan</i>
<i>human</i>	<i>gorilla</i>	<i>chimpanzee</i>		

- Using your observations, which species appear to be most closely related?

Students responses will vary, however they may draw comparisons between chimpanzees, orang-utans and gorillas; and between humans and gibbons. While this may seem contrary to modern taxonomy, humans and gibbons appear to have similar anatomical features.

- Did you use a particular characteristic to decide on the relationships? Explain why this characteristic was more important than others; or if you didn't use a specific characteristic, explain your reasons.

Student responses will vary. Cranial capacity is an identifying feature that traditionally has been used for classifying primates. Canine size and dentition pattern may also be used, as they indicate a life style for a primate that reflects the environment in which it lives.

- Sexual dimorphism, or difference due to gender, is present in some of the primate species included in *Primate trends*. However, all images in the learning object are of masculine skulls.

Compare images of male and female *Pongo* skulls below, to answer questions 5 and 6.

5. Describe three features that demonstrate sexual dimorphism between male and female *Pongo*.
The male skull is more robust than the female.
The male has larger canines than the female.
The zygomatic arches of the male are more robust and flared.
The crests of the male skull are more pronounced.
6. Do any features of the female *Pongo* skull change the way you've described primate relationships in question 4? For example, does the female *Pongo* appear human-like?
Students may answer that if female specimens were compared fewer differences may be noted between species.
7. Rank skull characteristics from least to most useful in deciding who is related to whom.
Student responses will vary. A suggested order is: brow ridge, maxilla length, mandible size, canine length and cranial capacity.
8. We know that *Pan* (chimpanzee) and *Homo* (human) are most closely related of these primates. Did this exercise reflect that? Why or why not?
Chimpanzee and human skulls are very different in appearance. Anatomical features reflect selection pressures and those vary greatly between humans and chimpanzees.
9. In this case, we've used a skull to determine how five primates are related. What problems are associated with looking at only a single piece of evidence?
Modern taxonomies reflect use of multiple types of evidence when grouping animals. Anatomy, behaviour, fossils, vestigial organs, molecular evidence and embryology are all powerful tools that when used together provide the most up to date, overall picture of relationships. If only one of these is used relationships cannot be accurately determined.

Part 2: Virtual zoo

10. Observe primates at Perth Zoo, or use the *Virtual zoo* interaction. Record your observations of characteristics and behaviour, in the species' passports provided. Then collate this information using the table on page 9. Notes for humans have been completed for you.

Consider how each feature may be used to determine relatedness between these animals.

11. Has observation of living primates changed your view about who is most closely related to whom? Explain your answer.
Student responses will vary.

OBSERVED CHARACTERISTIC OR BEHAVIOUR					
	orang-utan	gibbon	chimpanzee	gorilla	human
ratio of arm to leg length	<i>arms longer than legs</i>	<i>arms longer than legs</i>	<i>arms longer than legs</i>	<i>arms longer than legs</i>	legs longer than arms
arboreal or terrestrial	<i>mainly arboreal</i>	<i>mainly arboreal</i>	<i>arboreal and terrestrial</i>	<i>mainly terrestrial</i>	terrestrial
thumb length and opposability	<i>opposable</i>	<i>opposable</i>	<i>opposable</i>	<i>opposable</i>	opposable
mode of locomotion	<i>brachiation in trees, knuckle-walking on ground</i>	<i>mainly brachiation in trees</i>	<i>brachiation in trees, knuckle-walking on ground</i>	<i>mainly knuckle-walking on ground</i>	bipedal
foot adaptations to locomotion style	<i>grasping toe, weight on sides of feet rolling weight forward onto fists</i>	<i>grasping toe</i>	<i>grasping toe, plantigrade</i>	<i>grasping toe, plantigrade</i>	transverse and longitudinal arch
social interactions	<i>caring for young, object manipulation</i>	<i>interaction between family members, social hierarchy in place</i>	<i>living in groups, caring for young, also older members rearing young, grooming</i>	<i>living in groups, caring for young</i>	living in groups, complex social rules
sexual dimorphism (eg colour)	<i>males larger (taller and heavier) than females, male skulls more robust than females, with crests for muscle attachment</i>	<i>minimal differences</i>	<i>males are larger (taller and heavier) than females male skulls more robust than females, with crests for muscle attachment</i>	<i>males larger (taller and heavier) than females, coat colour differences, male skulls more robust than females, with crests for muscle attachment</i>	minimal differences
object manipulation & tool use	<i>using stick to probe for food</i>	<i>unwrapping enrichment items</i>	<i>using stick to probe for termites</i>	<i>none in this video</i>	advanced tool user